AgroNex-Pi
Complete Hardware Pin Connections and Wiring Guide
Autonomous Agricultural Robot | Raspberry Pi 4B Platform

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1. Components Overview

#	Component	Interface	Primary Role	Voltage
1	Raspberry Pi 4 Model B (4 GB)	GPIO / I2C / UART / CSI	Main controller	5 V (USB-C)
2	ADS1115 -- 16-bit 4-ch ADC	I2C (0x48)	Read analog sensors	3.3 V
3	Capacitive Soil Moisture Sensor	ADS1115 A0	Soil water content (0-100%)	3.3-5 V
4	HW-038 Water Level Sensor (3-pin)	ADS1115 A1	Reservoir fill level	3.3 V ONLY
5	DHT11 Temperature and Humidity	GPIO 4 (one-wire)	Ambient air monitoring	3.3-5 V
6	HMC5883L 3-Axis Compass	I2C (0x1E)	Heading and navigation	3.3 V
7	NEO-7M GPS Module	UART (TXD0/RXD0)	Geo-location lat/lon	3.3-5 V
8	2-Ch 5V Relay Module	GPIO 17, 27	Pump and Cutter switching	5 V coil
9	Dual BTS7960 43A Motor Driver	GPIO 5,6,13,19	Drive 4 DC wheels	12 V motor / 5 V logic
10	Buzzer (Active, 5 V)	GPIO 22	Audio alerts and diagnostics	5 V
11	Alert LED	GPIO 23 (330 Ohm)	Visual status indicator	3.3 V GPIO
12	Picamera2 (CSI)	CSI ribbon port	Disease scanning and live stream	Pi-powered

2. Raspberry Pi 4B -- 40-Pin GPIO Reference

Board Pin	GPIO / Function	Connected To	Direction	Notes
Pin 1	3.3 V Power	ADS1115 VDD + HMC5883L VCC + DHT11 VCC	OUT (power)	Max ~50 mA total
Pin 2	5 V Power	Relay VCC, BTS7960 VCC logic, GPS VCC	OUT (power)	From PSU/buck
Pin 3	GPIO 2 (SDA1)	ADS1115 SDA + HMC5883L SDA	I2C data	Shared bus, 4.7k pull-up
Pin 5	GPIO 3 (SCL1)	ADS1115 SCL + HMC5883L SCL	I2C clock	Shared bus, 4.7k pull-up
Pin 6	GND	Common star ground	GND	All GND bus
Pin 7	GPIO 4	DHT11 DATA	IN (1-wire)	1 kOhm pull-up to 3.3 V
Pin 8	GPIO 14 (TXD0)	NEO-7M RX	OUT (UART TX)	Pi transmits to GPS
Pin 10	GPIO 15 (RXD0)	NEO-7M TX	IN (UART RX)	GPS NMEA sentences to Pi
Pin 11	GPIO 17	Relay IN1 (Pump)	OUT (digital)	HIGH = Pump ON
Pin 13	GPIO 27	Relay IN2 (Cutter)	OUT (digital)	HIGH = Cutter ON
Pin 15	GPIO 22	Buzzer (+)	OUT (digital)	Buzzer GND to rail

Board Pin	GPIO / Function	Connected To	Direction	Notes
Pin 16	GPIO 23	Alert LED (330 Ohm)	OUT (digital)	Cathode to GND
Pin 29	GPIO 5	BTS7960 Left RPWM	OUT (PWM)	Left side -- Forward
Pin 31	GPIO 6	BTS7960 Left LPWM	OUT (PWM)	Left side -- Backward
Pin 33	GPIO 13	BTS7960 Right RPWM	OUT (PWM)	Right side -- Forward
Pin 35	GPIO 19	BTS7960 Right LPWM	OUT (PWM)	Right side -- Backward
CSI	CSI ribbon port	Raspberry Pi Camera Module	IN (video)	Picamera2 library

3. ADS1115 -- 16-bit I2C ADC Wiring

ADS1115 Pin	Connects To	Notes
VDD	Pi Pin 1 (3.3 V)	Do NOT use 5 V
GND	Pi GND rail	Common ground
SCL	Pi Pin 5 (GPIO 3)	Shared I2C bus with HMC5883L
SDA	Pi Pin 3 (GPIO 2)	Shared I2C bus with HMC5883L
ADDR	GND	I2C address = 0x48 (ADDR->GND)
A0	Soil Moisture AOUT	0-3.3 V analog output (dry=high, wet=low)
A1	HW-038 Water Sensor AOUT	0-3.3 V analog output (higher=more water)
A2	Not connected	Spare channel for future sensor
A3	Not connected	Spare channel for future sensor

```
Calibration: ADS_SOIL_DRY=26500 (air), ADS_SOIL_WET=13000 (water). Moisture% =
clamp((DRY-raw)/(DRY-WET)*100, 0, 100)
```

4. Capacitive Soil Moisture Sensor (v1.2)

Sensor Pin	Connects To	Notes
VCC	Pi Pin 1 (3.3 V) or 5 V	Module works at both voltages
GND	Pi GND rail	Common ground
AOUT	ADS1115 A0	Dry=~3.0 V, Wet=~1.0 V

5. HW-038 Water Level Sensor (3-Pin)

Sensor Pin	Connects To	Notes
VCC (+)	Pi Pin 1 (3.3 V)	3.3 V ONLY -- protects ADS1115 input
GND (-)	Pi GND rail	Common ground
AOUT (S)	ADS1115 A1	Voltage rises with water level (0-3.3 V)

```
HW038_RAW_EMPTY~0, HW038_RAW_FULL~25000. Level% = clamp(raw/25000*100, 0, 100). WATER_CRITICAL_LOW=15%
```

6. DHT11 -- Temperature and Humidity Sensor

DHT11 Pin	Connects To	Notes
Pin 1 (VCC)	Pi Pin 1 (3.3 V)	Also works at 5 V
Pin 2 (DATA)	Pi Pin 7 (GPIO 4)	1 kOhm pull-up resistor to 3.3 V
Pin 3 (NC)	Not connected	Leave open
Pin 4 (GND)	Pi GND rail	Common ground

```
Library: Adafruit_DHT | Sampling: every 30 seconds | Max rate: 1 Hz
```

7. HMC5883L -- 3-Axis Digital Compass (I2C 0x1E)

Compass Pin	Connects To	Notes
VCC	Pi Pin 1 (3.3 V)	3.3 V ONLY -- 5 V damages chip
GND	Pi GND rail	Common ground
SCL	Pi Pin 5 (GPIO 3 / SCL1)	Shared I2C bus (0x1E vs ADS1115 0x48 -- no conflict)
SDA	Pi Pin 3 (GPIO 2 / SDA1)	Shared I2C bus
DRDY	Not connected	Polling mode used in firmware

`heading = (atan2(Y,X) * 180/pi + 360) % 360 | i2cdetect must show 0x1E`

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8. NEO-7M GPS Module

WARNING: NEO-7M outputs 3.3 V UART. Do NOT connect to 5 V UART without a level shifter.

GPS Pin	Connects To	Notes
VCC	Pi Pin 2 (5 V)	Module has onboard 3.3 V regulator
GND	Pi GND rail	Common ground
TX	Pi Pin 10 (GPIO 15 / RXD0)	GPS NMEA data to Pi UART RX
RX	Pi Pin 8 (GPIO 14 / TXD0)	Pi UART TX to GPS (optional, for config)
PPS	Not connected	1 PPS -- reserved for future time sync use

`add enable_uart=1 to /boot/config.txt | Baud: 9600 | Library: pynmea2 | parse GGA/RMC sentences`

9. 2-Channel 5 V Relay Module

Relay Terminal	Connects To	Notes
VCC (module)	Pi Pin 2 (5 V)	Relay coil power
GND (module)	Pi GND rail	Common ground
IN1	Pi Pin 11 (GPIO 17)	Pump relay trigger -- HIGH = ON
IN2	Pi Pin 13 (GPIO 27)	Cutter relay trigger -- HIGH = ON
Relay 1 COM	12 V + from battery	Common terminal
Relay 1 NO	Pump motor (+)	Normally Open -- closes when GPIO17 HIGH
Relay 2 COM	12 V + from battery	Common terminal
Relay 2 NO	Cutter motor (+)	Normally Open -- closes when GPIO27 HIGH

Flyback diode across each relay coil recommended for GPIO protection.

10. Dual BTS7960 43A H-Bridge Motor Drivers

BTS7960 Pin	Pi GPIO	Function	Notes
RPWM (Left)	GPIO 5 (Pin 29)	Left -- Forward PWM	0-100% duty cycle
LPWM (Left)	GPIO 6 (Pin 31)	Left -- Backward PWM	0-100% duty cycle
RPWM (Right)	GPIO 13 (Pin 33)	Right -- Forward PWM	0-100% duty cycle
LPWM (Right)	GPIO 19 (Pin 35)	Right -- Backward PWM	0-100% duty cycle
VCC (both)	Pi Pin 2 (5 V)	Logic supply	5 V logic rail
GND (both)	Pi GND + Battery GND	Common ground	Star ground point
B+ (both)	12 V Battery (+)	Motor power supply	10-27 V range
M+ / M- (both)	DC Hub motors	Motor output terminals	Check polarity
R_EN / L_EN	Tie HIGH to VCC (5V)	Enable pins	Permanent enable

11. Power Distribution Architecture

Rail	Source	Consumers	Max Current
12 V	12 V Battery (LiFePO4 or SLA)	BTS7960 motor power, Relay COM terminals	10-20 A burst
5 V	5 V Buck converter (from 12 V)	Raspberry Pi, Relay coils, BTS7960 logic	3-5 A
3.3 V	Pi onboard LDO	ADS1115, HMC5883L, DHT11, GPS, Sensors	<= 200 mA
GND	Star ground single point	All modules share common GND	--

Use a 12V to 5V buck converter rated >= 5 A (e.g. LM2596 or XL4016). Keep motor and logic GND paths short.

12. Buzzer and Alert LED

Component	GPIO	Board Pin	Wiring	Notes
Active Buzzer (5V)	GPIO 22	Pin 15	Buzzer(+) to GPIO22, Buzzer(-) to GND	Beeps on alerts/startup test

Component	GPIO	Board Pin	Wiring	Notes
LED (3 or 5 mm)	GPIO 23	Pin 16	Anode to 330 Ohm to GPIO23, Cathode to GND	Blinks in SAFE_STATE/E-stop

13. Raspberry Pi Camera (Picamera2 / CSI)

Step	Action
1	Connect Pi Camera ribbon to CSI port (blue side toward USB ports on Pi 4B), lock the clip.
2	raspi-config > Interface Options > Legacy Camera > Disable (Picamera2 does not need legacy mode)
3	Install: pip install picamera2
4	Test import: python3 -c 'from picamera2 import Picamera2; c=Picamera2(); c.start(); c.stop()'
5	Live stream endpoint: GET http://:5000/video_feed (MJPEG). Shown in dashboard Live Feed section.

14. I2C + UART Verification Commands

```
# Enable I2C:
sudo raspi-config # > Interface Options > I2C > Enable
sudo apt install -y i2c-tools
sudo i2cdetect -y 1
# Expected: 0x1E (HMC5883L), 0x48 (ADS1115)
# Enable UART for GPS:
sudo raspi-config # > Interface > Serial Port > login shell=NO, hardware=YES
sudo cat /dev/ttyAMA0 # should print $GPGGA NMEA sentences at 9600 baud
```

15. Pre-Power Checklist

[ ] All GND lines connected to common star-ground rail
[ ] ADS1115, HMC5883L, DHT11, GPS sensors all on 3.3 V -- NOT 5 V directly
[ ] I2C pull-up resistors (4.7 kOhm) present on SDA/SCL lines
[ ] i2cdetect -y 1 shows 0x48 (ADS1115) and 0x1E (HMC5883L)
[ ] DHT11 pull-up (1 kOhm) on DATA line
[ ] GPS serial output visible on /dev/ttyAMA0 at 9600 baud
[ ] Relay isolated -- 12 V motor circuit NOT connected until code tested
[ ] BTS7960: VCC=5 V logic; B+=12 V motor; motors connected last
[ ] Camera ribbon fully inserted and locked; Picamera2 import test passes
[ ] 5 V buck converter output measured 4.9-5.1 V before connecting Pi
[ ] Flask server running: python app.py -- dashboard accessible on :5000
[ ] pip install -r requirements-pi.txt completed without errors

16. Required Python Packages (requirements-pi.txt)

Package	Version	Purpose
flask	>=2.3	Web server and REST API for dashboard
flask-cors	>=4.0	Cross-origin requests from AWS dashboard
Adafruit-DHT	>=1.4	DHT11 temperature and humidity driver
smbus2	>=0.4	I2C communication -- ADS1115 and HMC5883L
pynmea2	>=1.19	NMEA sentence parser for NEO-7M GPS

Package	Version	Purpose
picamera2	>=0.3	Raspberry Pi camera MJPEG stream
RPi.GPIO	>=0.7	GPIO control -- relay, buzzer, LED, motor PWM
boto3	>=1.34	AWS S3 sync -- scan logs and telemetry relay
torch	>=2.0	PyTorch ML inference for plant disease detection
torchvision	>=0.15	Vision transforms and pre-trained ResNet models
Pillow	>=10.0	Image preprocessing for disease detection
opencv-python-headless	>=4.8	Leaf segmentation and image analysis
numpy	>=1.25	Numerical operations for sensor math and ML
pyserial	>=3.5	Serial UART interface for GPS module
requests	>=2.31	HTTP API calls to AWS Lambda and API Gateway

End of AgroNex-Pi Hardware Pin Connections Guide

Firmware files: config.py | hardware.py | main.py | app.py